

## In-Process QC Checklist for Stationary Kavach Installation

QC Engineer Name: sushma

v1.2

| Station ID | Station Name | Zone | Division                           | Section Name                       | Initial Date | Updated Date |
|------------|--------------|------|------------------------------------|------------------------------------|--------------|--------------|
| 67776      | yuyuy        | ECR  | Pt Deendayal Upadhy - Pradhankhnta | Pt Deendayal Upadhy - Pradhankhnta | 24/03/2026   | 24/03/2026   |

| Total Points | Closed Points | Open Points |
|--------------|---------------|-------------|
| 166          | 100           | 66          |

|                     |                                    |                                                   |
|---------------------|------------------------------------|---------------------------------------------------|
| Installation Status | <input type="checkbox"/> Completed | <input checked="" type="checkbox"/> Not Completed |
|---------------------|------------------------------------|---------------------------------------------------|

## 1.0 Serial Number Verification

| S.No | Units                        | Status   | Observation | Image |
|------|------------------------------|----------|-------------|-------|
| 1.1  | Stationary Kavach Unit       | Matching |             |       |
| 1.2  | Peripheral Processing Card 1 | Matching |             |       |
| 1.3  | Peripheral Processing Card 2 | Matching |             |       |
| 1.4  | Vital Computer Card -1       | Matching |             |       |
| 1.5  | Vital Computer Card -2       | Matching |             |       |
| 1.6  | Vital Computer Card -3       | Matching |             |       |
| 1.7  | Voter Card -1                | Matching |             |       |
| 1.8  | Voter Card -2                | Matching |             |       |
| 1.9  | Vital Gateway Card 1 (S2S)   | Matching |             |       |
| 1.10 | Vital Gateway Card 2 (S2S)   | Matching |             |       |
| 1.11 | Vital Gateway Card 3 (NMS)   | Matching |             |       |
| 1.12 | EI Gateway-1                 | Matching |             |       |
| 1.13 | EI Gateway-2                 | Matching |             |       |
| 1.14 | Field Scanner Card 1         | Matching |             |       |
| 1.15 | Field Scanner Card 2         | Matching |             |       |
| 1.16 | Field Scanner Card 3         | Matching |             |       |
| 1.17 | Field Scanner Card 4         | Matching |             |       |
| 1.18 | Field Scanner Card 5         | Matching |             |       |
| 1.19 | Field Scanner Card 6         | Matching |             |       |

| S.No | Units                             | Status   | Observation | Image |
|------|-----------------------------------|----------|-------------|-------|
| 1.20 | Field Scanner Card 7              | Matching |             |       |
| 1.21 | Field Scanner Card 8              | Matching |             |       |
| 1.22 | RIU communication card 1-Host     | Matching |             |       |
| 1.23 | RIU communication card 2-Host     | Matching |             |       |
| 1.24 | RS 232-OFC converter 1 (STATION)  | Matching |             |       |
| 1.25 | RS 232-OFC converter 2 (STATION)  | Matching |             |       |
| 1.26 | RS 485-OFC converter (STATION)    | Matching |             |       |
| 1.27 | FIU Termination Card 1            | Matching |             |       |
| 1.28 | FIU Termination Card 2            | Matching |             |       |
| 1.29 | FIU Termination Card 3            | Matching |             |       |
| 1.30 | FIU Termination Card 4            | Matching |             |       |
| 1.31 | FIU Termination Card 5            | Matching |             |       |
| 1.32 | FIU Termination Card 6            | Matching |             |       |
| 1.33 | FIU Termination Card 7            | Matching |             |       |
| 1.34 | FIU Termination Card 8            | Matching |             |       |
| 1.35 | DPS Card 1                        | Matching |             |       |
| 1.36 | DPS Card 2                        | Matching |             |       |
| 1.37 | GPS & GSM Antenna-1               | Matching |             |       |
| 1.38 | GPS & GSM Antenna-2               | Matching |             |       |
| 1.39 | SMOCIP Unit                       | Matching |             |       |
| 1.40 | SMOCIP Termination Panel          | Matching |             |       |
| 1.41 | Station Termination Panel         | Matching |             |       |
| 1.42 | Stationary PDU Box                | Matching |             |       |
| 1.43 | IPS PDU Box                       | Matching |             |       |
| 1.44 | DC-DC Converter (Relay racks)     | Matching |             |       |
| 1.45 | RTU-1                             | Matching |             |       |
| 1.46 | RTU-2                             | Matching |             |       |
| 1.47 | Station Radio Power Supply card-1 | Matching |             |       |
| 1.48 | Station Radio Power Supply card-2 | Matching |             |       |
| 1.49 | Next Gen/. Cal Amp Radio Modem-1  | Matching |             |       |
| 1.50 | Next Gen/. Cal Amp Radio Modem-2  | Matching |             |       |
| 1.51 | RS 232-OFC converter 1 (RTU-1)    | Matching |             |       |

| S.No | Units                           | Status   | Observation | Image |
|------|---------------------------------|----------|-------------|-------|
| 1.52 | RS 232-OFC converter 2 (RTU-2)  | Matching |             |       |
| 1.53 | RS 485-OFC converter (SM-OCIP)  | Matching |             |       |
| 1.54 | RIU                             | Matching |             |       |
| 1.55 | RIU Power Supply Card-1         | Matching |             |       |
| 1.56 | RIU Power Supply Card-2         | Matching |             |       |
| 1.57 | RIU communication card 1-Remote | Matching |             |       |
| 1.58 | RIU communication card 2-Remote | Matching |             |       |
| 1.59 | Field Scanner Card 1            | Matching |             |       |
| 1.60 | Field Scanner Card 2            | Matching |             |       |
| 1.61 | Field Scanner Card 3            | Matching |             |       |
| 1.62 | Field Scanner Card 4            | Matching |             |       |
| 1.63 | RIU Battery Charge Cum Filter-1 | Matching |             |       |
| 1.64 | RIU Battery Charge Cum Filter-2 | Matching |             |       |

## 2.0 Building

| S.No | Aspect                | Requirement                                                                                                                                         | Status    | Observation | Image |
|------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------|
| 2.1  | KAVACH Building Works | Building Commissioning Report from the contractor shall be available with the customer's sign-off, and there shall be no open points in the report. | Available |             |       |

## 3.0 Tower

| S.No | Aspect             | Requirement                                                                                                                                      | Status    | Observation | Image |
|------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------|
| 3.1  | Tower Installation | Tower Commissioning Report from the contractor shall be available with the customer's sign-off, and there shall be no open points in the report. | Available |             |       |

## 7.0 OFC Networking Rack

| S.No | Aspect                                     | Requirement                                                                                                               | Status | Observation | Image |
|------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 7.1  | Installation                               | Network rack shall be installed as per approved floor plan and installation shall be done as per diagram 5 16 76 0058.    | Ok     |             |       |
| 7.2  | Installation                               | Patch cord routing shall be neat & bend radius to be maintained.                                                          | Ok     |             |       |
| 7.3  | Cabling                                    | All external cables entering the network rack shall pass through the grommets.                                            | Ok     |             |       |
| 7.4  | Cabling                                    | OFC cables shall be marked using naming tie-tags for easy identification of default and standby links.                    | Ok     |             |       |
| 7.5  | Earthing                                   | All networking modules inside the rack shall be connected to the rack chassis using 2.5 sq.mm green/yellow earthing wire. | Ok     |             |       |
| 7.6  | Earthing                                   | Network rack shall be connected to ring earth using a 10 sq.mm green/yellow earthing wire.                                | Ok     |             |       |
| 7.7  | Earthing                                   | Bolts shall be tightened to a torque of 8 Nm, and torque marking shall be applied using yellow paint.                     | Ok     |             |       |
| 7.8  | FDMS Box Installation in Network Rack      | OFC cables shall be spliced in the FDMS box as per network drawing.                                                       | Ok     |             |       |
| 7.9  | FDMS Box Installation in Network Rack      | FDMS boxes shall be clearly marked to identify up and down links.                                                         | Ok     |             |       |
| 7.10 | OFC cable continuity check, after splicing | OTDR test reports shall be available.                                                                                     | Ok     |             |       |

## 8.0 Relay Rack

| S.No | Aspect                      | Requirement                                                                                                                                                                                   | Status | Observation | Image |
|------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 8.1  | Installation                | Relay rack shall be installed as per the approved Floor Plan drawing, mounted on insulators, and secured to the floor by grouting as per diagram 5 16 76 0059.                                | Ok     |             |       |
| 8.2  | Wiring                      | Labeling sleeve shall be used to identify wiring with rack number, row number, relay number & contact type, Labelling shall be provided to relay contact wires at FTC PCBA of Station Kavach. | Ok     |             |       |
| 8.3  | Wiring                      | For EI Stations, verify all connections as per the approved EI Interface diagrams (vendor specific).                                                                                          | Ok     |             |       |
| 8.4  | Wiring                      | WAGO terminal details shall be as per interface circuit diagram 5 16 49 0685.                                                                                                                 | Ok     |             |       |
| 8.5  | Continuity Test / Bell Test | Completed Station Analyser and Bell Test reports shall be available.                                                                                                                          | Ok     |             |       |
| 8.6  | Earthing                    | Unit shall be connected to the ring earth/busbar using a 10 sq.mm green/yellow earthing wire, and bolts shall be tightened to a torque of 8 Nm with torque marking applied.                   | Ok     |             |       |
| 8.7  | Earthing                    | Crimping of lugs on earthing cables shall be carried out, and self-vulcanizing utility tape shall be applied.                                                                                 | Ok     |             |       |

## 9.0 Earthing

| S.No | Aspect                       | Requirement                                                                                                                             | Status    | Observation | Image |
|------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------|
| 9.1  | Installation                 | Earthing shall be done as per RDSO Spec RDSO/SPN/197/2008                                                                               | Ok        |             |       |
| 9.2  | Earth Resistance Measurement | Earth resistance shall be measured using a calibrated earth resistance tester. The measured value shall be less than or equal to 1 Ohm. | Ok        |             |       |
| 9.3  | Test Reports                 | Earth resistance test reports shall be available                                                                                        | Ok        |             |       |
| 9.4  | Labeling                     | All earth pits, earthing conductors, and earth points shall be clearly labelled and identifiable.                                       | Available |             |       |

## 10.0 Indoor Wiring / Cabling

| S.No | Aspect                   | Requirement                                                                                                                       | Status | Observation | Image |
|------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 10.1 | Redundant cabling        | Cabling shall be done as per station connectivity diagram 5 16 49 0614<br>Segregation of power & communication cables.            | Ok     |             |       |
| 10.2 | SMOCIP                   | 12 Core Signaling cable shall be used for button, counter & power supply.<br>OFC armoured cables shall be used for communication. | Ok     |             |       |
| 10.3 | No Joints                | No joints permitted except inside junction boxes or panels.                                                                       | Ok     |             |       |
| 10.4 | Color Coding             | Colour codes shall be followed for phase, neutral, and earth.                                                                     | Ok     |             |       |
| 10.5 | Tagging/Labeling         | Cables shall be tagged at both ends.                                                                                              | Ok     |             |       |
| 10.6 | Termination & Connection | Proper lugs/ferrules shall be used and terminals tightened correctly.                                                             | Ok     |             |       |

## 11.0 Outdoor Cabling

| S.No | Aspect                                                       | Requirement                                                                             | Status | Observation | Image |
|------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------|--------|-------------|-------|
| 11.1 | Cable route plan for OFC & Power cable (Relay Room to Tower) | Railway-approved outdoor signalling cable route plan shall be available.                | Ok     |             |       |
| 11.2 | Cable route markers                                          | Route markers shall be installed and clearly visible.                                   | Ok     |             |       |
| 11.3 | Cable route markers                                          | Where contractually required, underground RFID markers shall be installed and verified. | Ok     |             |       |

## 12.0 RFID Tags

| S.No | Aspect                                              | Requirement                                                                                                                                                                                                  | Status    | Observation | Image |
|------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------------|-------|
| 12.1 | RFID Tag Installation                               | Ensure the main RFID tag is installed in the direction specified in the RFID tag layout, and a duplicate RFID tag shall be installed at a distance of 3–5 meters from the main tag, except for turnout tags. | Ok        |             |       |
| 12.2 | RFID Tag Installation                               | RFID tag mounting shall be firm and secure, ensuring no gap or mechanical play as per RFID tag installation procedure 5 53 76 0055.                                                                          | Ok        |             |       |
| 12.3 | RFID Tag data Validation and Placement verification | System-generated RFID data validation report and Placement verification report shall be available.                                                                                                           | Available |             |       |

### 13.0 Components Inspection

| S.No | Aspect                                                              | Requirement                                                                                                                                       | Status | Observation | Image |
|------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|-------|
| 13.1 | Materials Inspection (Which are not Inspected by RDSO or Consignee) | Verify that all subcontractors and HBL personnel use only approved make and part numbers as per the List of Acceptable I&C Materials EG-EN-FT-34. | Ok     |             |       |

The following are the pending items to be verified during the next inspection

| Section Name                                          | Pending Points                                                                                                                                                                       |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Section 4: Power Supply</b>                        | 4.1.1, 4.1.2, 4.1.3, 4.1.4, 4.1.5, 4.1.6, 4.1.7, 4.1.8, 4.2.1, 4.2.2, 4.2.3, 4.2.4, 4.2.5                                                                                            |
| <b>Section 5: Kavach Equipment</b>                    | 5.1.1, 5.1.2, 5.1.3, 5.1.4, 5.1.5, 5.1.6, 5.2.1, 5.2.2, 5.2.3, 5.2.4, 5.2.5, 5.2.6, 5.2.7, 5.2.8, 5.2.9, 5.3.1, 5.3.2, 5.3.3, 5.3.4, 5.3.5, 5.4.1, 5.4.2, 5.4.3, 5.4.4, 5.4.5, 5.4.6 |
| <b>Section 6: RF Communication equipment on tower</b> | 6.1.1, 6.1.2, 6.1.3, 6.1.4, 6.1.5, 6.1.6, 6.1.7, 6.1.8, 6.1.9, 6.1.10, 6.2.1, 6.2.2                                                                                                  |

**Note : This QC report is generated by the system and does not need a physical signature.**